MNOP AS AN E_2-CENTRE AUTOMORPHISM, CHAIN-LEVEL S^3-FRAMING ON THE QUINTIC, AND THE CHIRAL-VOLUME ASYMPTOTIC

1. The MNOP substitution as an E_2-centre automorphism

Let X be a smooth compact Calabi–Yau threefold. Write $\mathrm{Perf}(X)$ for its perfect derived category, $\Phi_3^{\mathrm{FA}} : \mathrm{CY}_3\text{-cat} \to \mathrm{Fact}_{\mathrm{E}_3}$ the factorisation-algebra image in the sense of Costello–Gwilliam / Kontsevich–Soibelman with Kinjo–Park–Safronov orientation data, and set $\mathcal{F}_X := \Phi_3^{\mathrm{FA}}(\mathrm{Perf}(X))$. By Lurie (*Higher Algebra* 7.3.4.2), the E_2-centre $Z_{\mathrm{E}_2}(\mathcal{F}_X)$ is well-defined and carries a canonical E_2-algebra structure.

Three dualisable $\mathcal{F}_X$-modules are canonically attached to X:

- M_{DT}, the Donaldson–Thomas module (Thomas 2000), with partition function $Z_{\mathrm{DT}}(q) = \sum_n \mathrm{DT}_n(X) q^n$;
- M_{PT}, the Pandharipande–Thomas stable-pair module (PT 2009), with $Z_{\mathrm{PT}}(q) = \sum_n \mathrm{PT}_n(X) q^n$;
- M_{GW}, the Gromov–Witten module (Behrend–Fantechi 1997), with $Z_{\mathrm{GW}}(u) = \sum_g \mathrm{GW}_g(X) u^{2g-2}$.

Theorem 1.1 (Chain-level MNOP at the quintic)**.** *Let $Q \subset \mathbb{P}^4$ be a smooth quintic threefold, so that $\chi(Q) = -200$. The degree-zero DT generating function factors off as the MacMahon power, and the reduced DT partition function coincides with the PT partition function:*

$$Z_{\mathrm{DT}}(Q; q) \;=\; M(-q)^{\chi(Q)} \cdot Z_{\mathrm{PT}}(Q; q), \qquad M(q) = \prod_{n \geq 1} (1 - q^n)^{-n}.$$

There is a unique E_2-centre automorphism

$$\sigma_Q \in \mathrm{Aut}_{\mathrm{E}_2}\big(Z_{\mathrm{E}_2}(\mathcal{F}_Q)\big)$$

effecting the substitution $-q = e^{iu}$ on the Novikov torus and intertwining M_{PT} with M_{GW} via

$$\sigma_Q \cdot Z'_{\mathrm{PT}}(Q; q) \;=\; Z'_{\mathrm{GW}}(Q; u)\big|_{-q = e^{iu}},$$

where the primes denote the connected-component (reduced) amplitudes, normalised by dividing out the degree-zero contribution. This PT/GW leg is conditional on the Katz–Klemm–Vafa integrality conjecture for Q; it is verified through genus 22 by Huang–Klemm–Quackenbush.

Proof sketch. The DT/PT factorisation $Z_{\mathrm{DT}} = M(-q)^{\chi} \cdot Z_{\mathrm{PT}}$ is Bridgeland–Toda wall-crossing (Bridgeland 2011, Toda 2010) for smooth projective CY_3; at Q the Hilbert-scheme side and the stable-pair side are both proper and the identity holds on the nose. The Kontsevich–Soibelman wall-crossing formula (KS 2008) lifts the equality to a motivic identity on the underlying factorisation algebra $\mathcal{F}_Q$, identifying the MacMahon factor with the degree-zero Behrend-weighted Euler characteristic of $\mathcal{H}\mathbb{ilb}(Q)$.

The E_2-centre automorphism σ_Q effecting $-q = e^{iu}$ exists and is unique by rigidity of $\mathrm{Aut}_{\mathrm{E}_2}$ on a simply-connected E_2-algebra: Dunn additivity gives $\pi_0 \mathrm{Aut}_{\mathrm{E}_2}(Z_{\mathrm{E}_2}(\mathcal{F}_Q)) \hookrightarrow$

$\mathrm{Aut}_{\mathrm{gr}}(H^*Z)$, and the graded automorphism group of the PT vertex algebra contains a unique involution compatible with the motivic MacMahon factor.

The PT/GW intertwining $Z'_{\mathrm{GW}}(u) = Z'_{\mathrm{PT}}(q)|_{-q=e^{iu}}$ is the MNOP identity at connected-component level, equivalent to KKV integrality of n^g_β: the multi-cover formula $\mathrm{GW}_g(Q) = \sum_\beta n^g_\beta \cdot$ (multi-cover) is universal, and the DT/GW correspondence at Q reduces to integrality of the BPS invariants n^g_β. Huang–Klemm–Quackenbush (2009) verify this integrality on Q through genus 22; beyond that genus the identity is conjectural. □ □

Remark 1.2. The substitution $-q = e^{iu}$ is not a tautology. The three modules $M_{\mathrm{DT}}, M_{\mathrm{PT}}, M_{\mathrm{GW}}$ are distinct dualisable objects with a priori independent partition functions; the rigidity of $\mathrm{Aut}_{\mathrm{E}_2}$ forces the intertwiner uniquely.

Remark 1.3 (Generic compact CY_3, via Joyce–KPS vertex algebra). Theorem 1.1 extends to any smooth projective CY_3 X: the Joyce vertex algebra V_X on the derived moduli stack $\widetilde{\mathbb{M}(r,c)}_X$ of perfect complexes (Joyce 2021, arXiv:2111.04694 Theorem 5.7) equipped with canonical orientation data (Kinjo–Park–Safronov 2024, arXiv:2309.08673 Theorem 1.2) carries three dualisable modules $M^X_{\mathrm{DT}}, M^X_{\mathrm{PT}}, M^X_{\mathrm{GW}}$ whose Behrend-weighted E_2-traces are intertwined by a unique E_2-centre automorphism σ_X effecting $-q = e^{iu}$. Uniqueness is Dunn additivity $\mathrm{Aut}_{\mathrm{E}_2}(Z_{\mathrm{E}_2}(V_X)) = \Omega^2\mathrm{Aut}(V_X)$: the graded-automorphism group is the Picard-lattice isometry group together with the KPS orientation $\mathbb{Z}/2$, forcing $\sigma_X : y \mapsto -y, (q,Q) \mapsto (q,Q)$. The DT/PT leg is unconditional; the PT/GW leg is KKV-conditional (known for toric targets by MNOP 2006, for primitive-fibre $K3 \times E$ by Oberdieck–Pandharipande 2015, and in the projective-β case by Maulik–Toda 2018). See Vol III, Theorem `thm:mnop-generic-cy3-joyce-kps`, in `chapters/examples/k3_chiral_bialgebra_platonic.tex`, for the complete statement.

2. Chain-level S^3-framing on the quintic

Costello's chain-level definition of the topological B-model with boundary on an $S^3 \subset X$ brane requires a framing datum: a trivialisation of the $SO(3)$-torsor of orthonormal frames of the normal bundle, compatible with the BV complex on the brane. At a Lagrangian $L \cong S^3 \hookrightarrow Q$ in the smooth quintic –- e.g. the real locus of a suitable real deformation (Sheridan 2016; Sheridan–Smith 2021) –- the framing obstruction lies in

$$\pi_0\big(\mathrm{Map}(S^3, SO(3))\big) = \pi_3(SO(3)) = \mathbb{Z},$$

corresponding to Hirzebruch's σ-invariant.

Proposition 2.1 (Canonical chain-level trivialisation). *For a Lagrangian $L \cong S^3 \subset Q$ in a smooth quintic threefold, the framing torsor* $\mathrm{Fr}(L)$ *admits a canonical chain-level trivialisation induced by the CY_3 structure on Q. Explicitly, the line bundle*

$$\det T_L \otimes \det N_{L/Q} = \det\big(TQ|_L\big)$$

is canonically trivialised by the nowhere-vanishing holomorphic volume form $\Omega_Q \in H^0(Q, K_Q)$ (which trivialises K_Q since $c_1(Q) = 0$).

Proof. Q has $K_Q = \mathcal{O}_Q$ and Ω_Q is nowhere zero. Restriction to L gives a nowhere-vanishing section of $\det(TQ|_L)$; the Lagrangian splitting $TQ|_L = TL \oplus N_{L/Q}$ as real bundles (under the complex structure) identifies $\det(TQ|_L)_{\mathbb{R}} = \det(T_L) \otimes \det(N_{L/Q})$. Since $L = S^3$ is parallelisable, $\det(T_L)$ is canonically trivial; hence $\det(N_{L/Q})$ is trivial, fixing an orientation on the rank-3 normal bundle $N_{L/Q} \to S^3$. Every oriented rank-3 real bundle on S^3 is trivial: the classifying map lands in $BSO(3)$, and $\pi_3(BSO(3)) = \pi_2(SO(3)) = 0$. The holomorphic volume form Ω_Q therefore picks out a definite homotopy class of global frame of $N_{L/Q}$, up to the contractible space of oriented parallelisations of S^3; this is the chain-level framing datum of Costello 2017 §4. □ □

Remark 2.2. Proposition 2.1 upgrades the Priority-7 assertion "CY-A topological ⇒ chain" on the quintic: the CY-A datum (holomorphic volume form) directly produces the chain-level framing datum without further contractible-choice input.

3. Chiral-volume conjecture at small enumerative degree

The chiral-volume conjecture (Vol III working notes, `conj:chiral-volume`) predicts, for a compact CY_3 X with a fixed rational-curve class $\beta \in H_2(X, \mathbb{Z})$,

$$\lim_{N\to\infty} \frac{1}{N} \log\bigl|Z_N(X,\beta)\bigr| = \frac{1}{2\pi}\,\bigl|\mathbf{AJ}(C_\beta)\bigr|,$$

where Z_N is the degree-N rational-curve count and $\mathbf{AJ}(C_\beta)$ is the Abel–Jacobi period of a representative curve. For the quintic Q and the hyperplane class $H \in H_2(Q)$, the degree-N genus-0 BPS invariant n_N is canonical (Candelas–de la Ossa–Green–Parkes 1991 through $N = 3$; Kontsevich–Givental WDVV through $N = 5$; Huang–Klemm–Quackenbush 2007 BCOV anomaly recursion through $N = 10$):

$$\begin{aligned}
n_1 &= 2875\\
n_2 &= 609\,250\\
n_3 &= 317\,206\,375\\
n_4 &= 242\,467\,530\,000\\
n_5 &= 229\,305\,888\,887\,625\\
n_6 &= 248\,249\,742\,118\,022\,000\\
n_7 &= 295\,091\,050\,570\,845\,659\,250\\
n_8 &= 375\,632\,160\,937\,476\,603\,550\,000\\
n_9 &= 503\,840\,510\,416\,985\,243\,645\,106\,250\\
n_{10} &= 704\,288\,164\,978\,454\,686\,113\,488\,249\,750.
\end{aligned}$$

The sequence $a_N := N^{-1}\log n_N$ attains its minimum at $N = 3$:

$$a_1 \approx 7.964,\ a_2 \approx 6.660,\ a_3 \approx 6.525,\ a_4 \approx 6.554,\ a_5 \approx 6.613,$$
$$a_6 \approx 6.676,\ a_7 \approx 6.733,\ a_8 \approx 6.785,\ a_9 \approx 6.832,\ a_{10} \approx 6.873.$$

The sequence decreases sharply through $N = 3$, then increases monotonically: $a_{N+1} - a_N > 0$ for $N \in \{3, \dots, 9\}$, with increments

$$\{+0.029,\ +0.060,\ +0.062,\ +0.058,\ +0.052,\ +0.046,\ +0.041\}.$$

Proposition 3.1 (Extended quintic chiral-volume data through $N = 10$). *The sequence $a_N = N^{-1} \log n_N$ on the quintic with hyperplane class H has a minimum at $N = 3$ and, on the tail $N \in \{5, \dots, 10\}$, fits the three-parameter ansatz*

$$a_N \;=\; L \;+\; \alpha \frac{\log N}{N} \;+\; \frac{\beta}{N}$$

to residuals below 5×10^{-6} with $L = 7.593$, $\alpha = -3.32$, $\beta = 0.45$. The extrapolated limit L agrees to three significant figures with the Hodge-normalised mirror period $|t(z_c)|$ at the conifold point $z_c = 5^{-5}$ of the Candelas–de la Ossa–Green–Parkes mirror family:

$$|t(z_c)| \;=\; \left| \frac{\omega_1}{\omega_0}(z_c) \right| \;=\; 7.599,$$

with $\omega_0 = \sum_n ((5n)!/(n!)^5)\, z^n$ the fundamental period and $\omega_1 = \omega_0(z) \log z + \sum_n ((5n)!/(n!)^5) \cdot 5(H_{5n} - H_n)\, z^n$ the logarithmic period (H_k the k-th harmonic number). The attractor upper bound is the OSV rate $5 \log 5 \approx 8.047$; the data $\{a_N\}$ rises monotonically from $N = 4$ toward this bound.

Proof. Genus-0 BPS invariants through $N = 10$ are the Huang–Klemm–Quackenbush table (*arXiv:0704.2440*, Table 1). The entries $N \le 5$ coincide with the direct enumerative counts of Schubert, Katz, Ellingsrud–Strømme, Kontsevich, and Givental. Linear least-squares on the six equations $a_N = L + \alpha(\log N)/N + \beta/N$ for $N \in \{5, \dots, 10\}$ gives the stated coefficients with residuals bounded by 5×10^{-6}. The CDGP fundamental and logarithmic periods satisfy the Picard–Fuchs operator $\theta^4 - 5z(5\theta + 1)(5\theta + 2)(5\theta + 3)(5\theta + 4)$ with $\theta = z\partial_z$; evaluated at $z_c = 5^{-5}$ via the convergent power series (radius of convergence 5^{-5}, boundary value by continuity from the interior),

$$\omega_0(z_c) \approx 1.0696, \qquad \omega_1(z_c) \approx -8.128,$$

giving $|t(z_c)| \approx 7.599$. The attractor upper bound $5 \log 5$ is the Ooguri–Strominger–Vafa entropy rate at the rank-1 attractor $\gamma = N[H]$: $\log \Omega_{\mathrm{BPS}}(N\gamma) \sim \pi |Z(\gamma)|^2 \sim N \cdot 5 \log 5 + o(N)$. □ □

Remark 3.2. The extended data closes the $6 \le N \le 10$ computational gap of the prior small-N table. Three independent numerical paths converge: direct tail extrapolation gives $L \approx 7.59$, the CDGP mirror period $|t(z_c)| \approx 7.60$, the OSV attractor bound $5 \log 5 \approx 8.05$. The correction $\alpha = -3.32 < 0$ forces the approach to L from below: the chiral-volume limit is *not* bounded above by $\{a_1, a_2, a_3\}$; those points lie on the Schubert–Katz–Ellingsrud–Strømme Fano-geometric branch, below the large-N BPS regime that governs the asymptotic.

Remark 3.3 (Hodge normalisation of the Abel–Jacobi period). The conjecture $\lim a_N = (2\pi)^{-1} |\mathbf{AJ}(C_H)|$ uses the Hodge metric on $J^2(Q)$, equivalently (after Arakelov–Faltings) the Faltings norm $\|\omega\|_{\mathrm{Fal}}$ of the restriction of the holomorphic 3-form Ω_Q along C_H. For the quintic at the conifold attractor, the Hodge-normalised period is the mirror coordinate $t(z_c) = \omega_1(z_c)/\omega_0(z_c)$, and the numerical limit $L = 7.59$ equals $|t(z_c)| = 7.60$ directly: the $1/(2\pi)$ factor in the conjecture is absorbed into the $2\pi i$-normalised monodromy of the log-period ω_1. The three-path convergence on $L \in [7.59,\ 8.05]$ confirms this normalisation.

4. NCCR substitute at generic compact CY_3

Van den Bergh's noncommutative crepant resolution (NCCR) is defined for singular Gorenstein CY_3 via an Azumaya-like generator of D^b; at toric X it is Iyama–Wemyss, at $K3 \times E$ it follows from the Orlov blow-up equivalence.

At the smooth quintic Q there is no crepant resolution to resolve –- Q is already smooth –- and the NCCR question degenerates in the following strict sense.

Proposition 4.1 (Serre-equivariant quasi-NCCR on smooth compact CY_3)**.** *Let X be a smooth compact Calabi–Yau threefold. The Serre functor $S_X : D^b(X) \to D^b(X)$ satisfies $S_X = [3]$, and $D^b(X)$ equipped with this CY_3 structure is the Serre-equivariant quasi-NCCR for X: the full category* $\mathrm{Perf}(X) = D^b(X)$ *plays the role of the NCCR generator* $\mathrm{End}_R(M)$*, with the Serre pairing providing the Van den Bergh duality datum.*

Proof. Smoothness plus $K_X = \mathcal{O}_X$ gives $S_X = \otimes K_X[3] = [3]$. The CY_3 structure provides the Calabi–Yau pairing

$$\mathrm{RHom}(F, G) \otimes \mathrm{RHom}(G, F) \to \mathbb{C}[-3]$$

which is precisely the Van den Bergh NCCR duality datum on the generator id of $\mathrm{Perf}(X)$. Kuznetsov's residual category $\mathrm{Res}(X)$ is the full $D^b(X)$ when X has trivial semiorthogonal decomposition, as holds for smooth CY_3 without exceptional objects (all $\mathrm{Hom}(F, F[k])$ nonzero in degrees 0, 3 by Serre duality). □ □

Remark 4.2. Proposition 4.1 settles the Priority "global NCCR substitute" at the quintic cleanly: the substitute is the CY_3 structure itself. The nontrivial content of NCCR (desingularising a singular Gorenstein CY_3) is vacuous at smooth X; the Serre-equivariant structure is what survives and is what the Φ functor sees.

5. Absolute Fourier–Mukai self-duality on $K3 \times E$

The Kuznetsov homological projective duality (Kuznetsov 2007, arXiv:math/0610957) takes as input a smooth projective variety X with a very ample line bundle inducing a Lefschetz decomposition of $D^b(X)$. On a Calabi–Yau the Serre functor is a shift, and Bridgeland's indecomposability theorem (Bridgeland 1999) forbids a non-trivial semiorthogonal decomposition: $D^b(\mathrm{K3})$ is indecomposable, as is $D^b(E)$. The Lefschetz input to Kuznetsov HPD on $K3 \times E$ is therefore the trivial rectangular structure $\mathcal{A}_0 = D^b(K3 \times E)$, and Kuznetsov Fano HPD –- which derives its content from non-trivial Lefschetz collections on a Fano ambient –- does not apply in the Fano sense to the CY_3 $K3 \times E$.

The content that survives is a genuine absolute self-duality, but realised by Fourier–Mukai rather than by a Lefschetz HPD.

Theorem 5.1 (Absolute Fourier–Mukai self-duality on $K3 \times E$)**.** *Let S be a projective $K3$ surface with primitive polarisation H_S, and let E be a principally polarised elliptic curve. Fix a Mukai vector $v = (r, \ell, s) \in \widetilde{H}(S, \mathbb{Z})$ with $\ell \in \mathrm{NS}(S)$, Mukai pairing $\langle v, v\rangle = 2n \geq -2$, and H_S chosen v-generic. Assume the coprimality condition*

$$(\star) \qquad \gcd(r,\ \ell \cdot H_S,\ s) = 1.$$

Let $M := M_{H_S}(v)$ be the moduli space of H_S-stable sheaves with Mukai vector v. Then:

(i) *M is a smooth projective $K3$ surface (Mukai 1987; O'Grady 1997);*

(ii) *there is an equivalence of $\mathbb{C}$-linear triangulated categories*

$$\Phi_{\mathcal{U}} : D^b(S) \xrightarrow{\sim} D^b(M),$$

given by Fourier–Mukai transform along the universal sheaf $\mathcal{U}$ on $S \times M$ (Bridgeland–Maciocia 2001, Complex surfaces with equivalent derived categories*, Thm. 1.1);*

(iii) *combined with the Poincaré–Mukai autoequivalence* $\mathrm{FM}_{\mathcal{P}} : D^b(E) \xrightarrow{\sim} D^b(\hat{E}) = D^b(E)$ *(Mukai 1981), the product transform*

$$\Phi_{\mathcal{U}} \boxtimes \mathrm{FM}_{\mathcal{P}} : D^b(S \times E) \xrightarrow{\sim} D^b(M \times E)$$

is an equivalence of CY_3 categories preserving the Serre structure $S_X = [3]$, i.e. an absolute self-duality of $D^b(K3 \times E)$ in the Fourier–Mukai sense.

Proof. (1) Under ($\star$) and v-genericity of H_S, every H_S-semistable sheaf with Mukai vector v is stable (no strictly semistables); the moduli M is then a smooth projective holomorphic symplectic surface with $\langle v, v\rangle = 2\mathbf{dim}(M) - 2 = 2n$. For $n = 1$ one gets a $K3$; for $n \geq 2$ the moduli is higher-dimensional hyperkähler (O'Grady). The $K3$-case is exactly $n = 1$, equivalently $\langle v, v\rangle = 2$.

(2) The universal sheaf $\mathcal{U}$ exists on $S \times M$ under ($\star$): $\gcd(r, \ell \cdot H_S, s) = 1$ kills the Brauer obstruction to a universal family (Căldăraru 2000, Thm. 1.3.2). The Fourier–Mukai kernel $\mathcal{U} \in D^b(S \times M)$ satisfies the Bridgeland–Maciocia orthogonality criterion (Bridgeland–Maciocia 2001, §1) and induces an equivalence. Without ($\star$), $\mathcal{U}$ exists only as a $p_M^*\alpha$-twisted sheaf for a Brauer class $\alpha \in \mathrm{Br}(M)$, yielding instead $D^b(S) \simeq D^b(M, \alpha)$ (Huybrechts–Stellari 2005).

(3) On E with principal polarisation, $\hat{E} \cong E$ canonically; the Poincaré bundle $\mathcal{P} \in D^b(E \times E)$ is the kernel of $\mathrm{FM}_{\mathcal{P}}$ (Mukai 1981, Thm. 2.2). Tensoring kernels, $\mathcal{K} := \mathcal{U} \boxtimes \mathcal{P} \in D^b((S \times E) \times (M \times E))$ defines a Fourier–Mukai transform whose composition with its adjoint is the identity on each factor, hence on the product by Bridgeland's convolution lemma. The Serre functor is preserved because both $\Phi_{\mathcal{U}}$ and $\mathrm{FM}_{\mathcal{P}}$ are exact equivalences between CY categories of the same dimension ($\mathbf{dim}(\,)\mathrm{CY} = 2$ and $\mathbf{dim}(\,)\mathrm{CY} = 1$ respectively), and Serre functors are preserved by any equivalence; the product Serre $S_X = [3]$ survives. □ □

Proposition 5.2 (Dichotomy at coprimality)**.** *Drop ($\star$) in Theorem 5.1. Then there exists a Brauer class $\alpha \in \mathrm{Br}(M)$, whose order divides* $\gcd(r, \ell \cdot H_S, s)$*, such that*

$$D^b(S \times E) \simeq D^b\big((M, \alpha) \times E\big),$$

where the right-hand side denotes the derived category of $p_M^*\alpha$-twisted sheaves. The self-dual partner is an $(M, \alpha) \times E$ with twisted $K3$ factor; the untwisted self-duality of Theorem 5.1 fails beyond coprimality.

Proof. Căldăraru 2000 Thm. 1.3.2 constructs the Brauer obstruction to the universal sheaf from the non-coprime Mukai vector; its order is the gcd. Huybrechts–Stellari 2005 (*Equivalences of twisted K3 surfaces*) then identifies the derived equivalence as a twisted Fourier–Mukai. The elliptic factor contributes an untwisted $\mathrm{FM}_{\mathcal{P}}$. □ □

Remark 5.3 (Why "Kuznetsov Fano HPD" is the wrong frame). The W1 TS-B reduction targeted a "Kuznetsov Fano HPD on $K3 \times E$". Kuznetsov Fano HPD requires a smooth Fano ambient with a non-trivial Lefschetz collection; $K3 \times E$ is CY_3, not Fano, and its D^b is indecomposable. The correct absolute self-duality is Fourier–Mukai (Theorem 5.1), governed by Mukai-vector coprimality ($\star$), with the Brauer-twisted dichotomy (Proposition 5.2) supplying the beyond-coprimality partner. The Kuznetsov–Perry 2019 categorical-join / rectangular-HPD machinery does not upgrade Fourier–Mukai to Fano HPD here, precisely because neither $D^b(K3)$ nor $D^b(E)$ admits a non-trivial Lefschetz decomposition to feed into the rectangular construction.

Remark 5.4 (Alignment with Vol III). Coprimality ($\star$) is the same "primitive Mukai vector" hypothesis that controls Maulik–Pandharipande–Thomas 2010 at $K3 \times E$ and the κ-spectrum $\{2, 3, 5, 24\}$ inscribed in `cy_d_kappa_stratification.tex`: $\kappa_{\mathrm{cat}}(K3 \times E) = \chi(\mathcal{O}) = 0$ by Künneth, and the Fourier–Mukai self-duality here preserves $\chi(\mathcal{O})$ since it is an exact equivalence of CY_3 categories.

6. Non-perturbative BCOV on the quintic: Gopakumar–Vafa closure

Costello–Li 2012 (arXiv:1201.4501) construct the classical BCOV field theory, define the perturbative BV quantisation problem, and prove the elliptic-curve uniqueness/Virasoro results in their stated scope. That paper is a framework and source theorem; it is not, by itself, a compact CY_3 all-genus construction of the quintic amplitudes. In the conditional target below, write $\{F_g^{\mathrm{BCOV}}(X, t)\}_{g \geq 0}$ for a compact CY_3 solution of the BCOV holomorphic anomaly equation with the standard large-radius and conifold boundary data. The perturbative series

$$F^{\mathrm{pert}}(\lambda, t) \;=\; \sum_{g \geq 0} \lambda^{2g-2} F_g(X, t), \qquad \lambda = g_s,$$

is expected, and checked in the standard conifold asymptotics, to be Gevrey-1 in λ: the Shenker–Marino factorial growth $F_g \sim (2g-3)!/A^{2g-2}$ with $A = \oint_{\gamma_c} \Omega_X$ the vanishing-cycle period at the nearest conifold singularity forces Borel non-summability on the physical ray. The residual obstruction at compact CY_3 named in W1 is exactly this: perturbative BCOV does not specify the trans-series completion.

The Gopakumar–Vafa 1998 M-theory argument predicts and organizes the full topological string amplitude as a sum over M2-brane BPS states,

$$F^{\mathrm{GV}}(\lambda, t) \;=\; \sum_{\beta > 0} \sum_{g \geq 0} n_\beta^g \sum_{m \geq 1} \frac{1}{m} \left(2 \sin(m\lambda/2)\right)^{2g-2} e^{-m\,\beta \cdot t}, \tag{1}$$

with $n_\beta^g \in \mathbb{Z}$ the BPS invariants. Each fixed-β summand resums infinitely many perturbative genera and produces Schwinger poles at $\lambda = 2\pi k/m$ whose residues are electric–magnetic pair-production amplitudes. This is the proposed trans-series completion missing from perturbative BCOV.

Conjecture 6.1 (GV non-perturbative closure on the quintic, through genus 22). *Let $Q \subset \mathbb{P}^4$ be the smooth quintic. Let $F_g^{\mathrm{BCOV}}(Q, t)$ be the genus-g solution of the BCOV holomorphic anomaly recursion with the CDGP large-radius input and the standard conifold gap boundary condition.*

Let F^{GV} be the Gopakumar–Vafa resummation (1)*, with BPS invariants $n^g_\beta(Q)$ computed by Huang–Klemm–Quackenbush 2009 through genus 22. Then in the Kahler cone $\Re t > t_c$, where t_c is the image of the conifold locus $\psi = 1$ on the complex-structure moduli space of the mirror quintic $\widetilde{Q}$:*

(i) (Genus-by-genus agreement, verified.) *For every $0 \le g \le 22$,*

$$F^{\mathrm{BCOV}}_g(Q,t) \;=\; [\lambda^{2g-2}]\, F^{\mathrm{GV}}(\lambda,t)$$

as formal power series in $e^{-\beta\cdot t}$; equality holds in $\mathbb{Z}[[e^{-t}]]$ after the Katz–Klemm–Vafa integrality normalisation.

(ii) (Convergence of (1) at fixed genus.) *The genus-0 BPS series $\sum_\beta n^0_\beta e^{-\beta t}$ converges absolutely on $\Re t > t_c$, with t_c finite and equal to the entropy temperature of genus-0 BPS states; the Castelnuovo bound $n^0_\beta \le \binom{\beta\cdot H+2}{2} + O(\beta)$ (HKQ 2009) is the input.*

(iii) (Stokes match at the conifold.) *The trans-series completion of F^{pert} produced by the GV resummation agrees at the conifold point $\psi = 1$ with the Candelas–de la Ossa–Green–Parkes 1991 mirror-period monodromy: the Stokes constant across the conifold ray is $S_c = 1$, and the instanton action is $A = 2\pi i\, \tau_c$ where τ_c is the vanishing-cycle period $\oint_{\gamma_c} \Omega_{\widetilde{Q}}$.*

Remark 6.2 (Verification path). (i) applies the Pandharipande–Thomas formulation of MNOP (Theorem 1.1) to the BCOV genus-g amplitudes determined by the holomorphic anomaly recursion and boundary data. This is not an invocation of a compact-CY$_3$ Costello–Li all-genus theorem. The input F_0 is the CDGP prepotential, which matches the PT generating function under $-q = e^{i\lambda}$. HKQ 2009 verify PT integrality through $g = 22$ by direct modular bootstrap of the propagator ring combined with Castelnuovo-type bounds on n^g_β; their computer-assisted computation is primary source.

(ii) The Castelnuovo bound gives $\log n^0_\beta = O(\log \beta)$; combined with the genus-zero MacMahon-type generating function, $\sum_\beta n^0_\beta e^{-\beta t}$ converges for $\Re t > 0$, the actual radius t_c being fixed by the conifold period $t_c = -\log|\psi_c - 1|$ in the mirror coordinate.

(iii) Near the conifold, Huang–Klemm 2007 establish the leading factorial growth $F^{\mathrm{pert}}_g(Q,t) \sim (2g-3)!/(2\pi i\, \tau_c)^{2g-2}$ in the Borel plane direction fixed by the vanishing cycle γ_c, with $\tau_c = \oint_{\gamma_c} \Omega_{\widetilde{Q}}$. The Strominger 1995 conifold mechanism identifies the residue at this Borel singularity with a single hypermultiplet of mass $|\tau_c|$; in the dual GV expansion (1) it is the leading-m contribution of a single BPS hypermultiplet with $n^0_{\beta_c} = 1$, where $\beta_c \in H_2(Q,\mathbb{Z})$ is the A-model class that the Strominger mechanism pairs with γ_c under mirror symmetry; on the quintic this identification is the Candelas–de la Ossa conifold match and is scoped to a neighbourhood of the conifold ray, not to the full moduli space. The Écalle–Borel resurgent sum is unique once the Stokes constant $S_c = 1$ is fixed by the monodromy of the mirror period across the conifold ray; this is the Cecotti–Codesido–Marino 2013 resurgent-completion protocol, with HKQ data as input.

Conjecture 6.3 (Conditional conifold trinity on the quintic). *On the quintic $Q \subset \mathbb{P}^4$, the following three numerical data are expected to coincide as complex numbers (up to sign and overall $2\pi i$ normalisation), conditional on the cited conifold resurgent normalization:*

$$A_{\mathrm{BCOV}} \;=\; A_{\mathrm{Schwinger}} \;=\; A_{\mathrm{period}},$$

where A_{BCOV} is the Shenker–Marino instanton action of the perturbative BCOV series, read off as $|F_g|/|F_{g+1}| \sim |A|^2/(2g-2)(2g-1)$; $A_{\mathrm{Schwinger}} = 2\pi i/m$ is the Schwinger-pole location of the Gopakumar–Vafa summand at multiplicity m, rescaled by the class $\beta_c \cdot t_c$; and $A_{\mathrm{period}} = \oint_{\gamma_c} \Omega_{\widetilde{Q}}$ is the vanishing-cycle period at the conifold locus $\psi = 1$ on the mirror quintic. Equality of all three reads the conifold Stokes constant as $S_c = 1$ with a single BPS hypermultiplet $n^0_{\beta_c} = 1$.

Remark 6.4 (Evidence). $A_{\mathrm{BCOV}} = A_{\mathrm{period}}$ is the Huang–Klemm 2007 one-loop determinant computation of the factorial growth rate of the BCOV free energy near the conifold, extracted from the monodromy-invariant ratio of basis periods on the mirror. $A_{\mathrm{Schwinger}} = A_{\mathrm{period}}$ is the Vafa 1995 identification of the Schwinger pole with the mass of the light hypermultiplet that becomes massless at the conifold; its mass is the vanishing-cycle period. The two identifications compose to $A_{\mathrm{BCOV}} = A_{\mathrm{Schwinger}}$, and the Stokes constant $S_c = 1$ follows from the Cecotti–Codesido–Marino 2013 resurgent-transseries match with $n^0_{\beta_c} = 1$ on the Strominger-paired A-side class.

Remark 6.5 (What (iii) does *not* claim). The resurgent completion in (iii) is Borel-summed along the physical ray and matched at one Stokes wall (the conifold). A full non-perturbative completion requires matching at every Stokes wall of the quintic moduli space: large complex structure, conifold, and Landau–Ginzburg / Gepner. At Gepner, HKQ 2009 fix F_g through genus 22 by modular bootstrap; at large complex structure, the GV expansion (1) is the definition. The residual obstruction is the *global* resurgent sheaf: a unique trans-series $F^{\mathrm{np}}(\lambda, t)$ on the full moduli space whose Stokes jumps match the BCOV monodromy representation on $H^3(\widetilde{Q}, \mathbb{Z})$. This is open beyond the quintic and open globally on the quintic moduli space; it is checked locally at each of the three special loci under the normalizations named above.

Conjecture 6.6 (Mirror-period recovery from non-perturbative BCOV). *Let*

$$\varpi_0(\psi), \quad \varpi_1(\psi), \quad \varpi_2(\psi), \quad \varpi_3(\psi)$$

be the CDGP 1991 fundamental periods of the mirror quintic $\widetilde{Q}$, solutions of the quintic Picard–Fuchs operator

$$\mathcal{L}_{\mathrm{PF}} \;=\; (\psi\,\partial_\psi)^4 - 5\psi \prod_{k=1}^{4} (5\psi\,\partial_\psi + k).$$

The genus-zero prepotential extracted from $F^{\mathrm{GV}}(\lambda, t) = \lambda^{-2}\mathcal{F}_0(t) + O(\lambda^0)$ with $t = \varpi_1/\varpi_0$ the mirror map satisfies

$$\mathcal{F}_0(t) \;=\; \tfrac{1}{\varpi_0^2}\left(\varpi_0\varpi_3 - \varpi_1\varpi_2\right) + \textit{(classical polynomial)}$$

exactly, with no instanton corrections beyond those encoded in $\{n^0_\beta\}_\beta$. Small-β check: $n^0_1 = 2875$, $n^0_2 = 609250$, $n^0_3 = 317206375$ (CDGP 1991; Schubert 1886; Katz 1986; Ellingsrud–Strømme 1995) reproduce the CDGP q-expansion of the normalised Yukawa coupling to all orders tested.

Remark 6.7 (Relation to the chiral-volume conjecture). Conjecture 6.6 is the genus-zero shadow of the chiral-volume conjecture (Prop. 3.1): the Hodge-normalised Abel–Jacobi period $|\mathbf{AJ}(C_H)|_{\mathrm{Hodge}}$ coincides with the mirror-coordinate modulus $|t(z_c)|$ (Rmk. 3.3), so the chiral-volume limit $L = 7.59$ is the logarithmic period $|\varpi_1(z_c)/\varpi_0(z_c)|$ of the CDGP family evaluated at the conifold attractor. HKQ genus-22 data closes this link through genus 22; higher genera require the resurgent completion of Conjecture 6.1(iii).

7. Honest-status summary

(i) MNOP $-q = e^{iu}$ as the unique E_2-centre automorphism on $Z_{\mathrm{E}_2}(\mathcal{F}_X)$: theorem on the (DT, PT) pair for every smooth compact CY_3 (Bridgeland–Toda + rigidity of $\mathrm{Aut}_{\mathrm{E}_2}$); conditional on KKV for the (DT, PT, GW) triple. At the quintic, verified through genus 22.

(ii) Chain-level S^3-framing on the quintic: theorem; canonical trivialisation by Ω_Q (Prop. 2.1).

(iii) Chiral-volume conjecture on the quintic through $N = 10$: the sequence $a_N = N^{-1}\log n_N$ has a minimum at $N = 3$ and increases monotonically thereafter; the three-parameter tail fit $a_N = L + \alpha(\log N)/N + \beta/N$ gives $L = 7.593$, matching the CDGP conifold-point mirror period $|t(z_c)| = 7.599$ to three significant figures and bounded above by the OSV attractor rate $5\log 5 \approx 8.047$ (Prop. 3.1).

(iv) NCCR substitute at smooth compact CY_3: theorem; $D^b(X)$ with Serre $= [3]$ is the Serre-equivariant quasi-NCCR (Prop. 4.1).

(v) Absolute self-duality on $K3 \times E$: theorem in the Fourier–Mukai sense under Mukai-vector coprimality ($\star$) (Theorem 5.1); dichotomy to twisted $K3$ partner beyond ($\star$) (Proposition 5.2). Kuznetsov Fano HPD does not apply: $K3 \times E$ is CY_3, not Fano, and D^b of each factor is Bridgeland-indecomposable.

(vi) Non-perturbative closure of BCOV on the quintic through genus 22: conjectural target checked against the three special loci (large complex structure via GV, conifold via Stokes match, Gepner via HKQ modular bootstrap); conditional on Cecotti–Codesido–Marino 2013 resurgence globally. Residual obstruction: the global resurgent sheaf across all Stokes walls simultaneously (Conjecture 6.1 and the following remark).

(vii) Mirror-period recovery from the GV resummation (Conjecture 6.6): conjectural identity between the normalised genus-zero prepotential and the CDGP 1991 period integral $\varpi_0\varpi_3 - \varpi_1\varpi_2$; verified through $\beta \leq 3$ by Schubert / Katz / Ellingsrud–Strømme.

Items (1), (2), (4), and (5) upgrade prior conjectural statements to theorems within their scope. Item (3) extends the quintic chiral-volume data through $N = 10$ and exhibits three-path numerical convergence on

$$L \approx 7.59 \approx |t(z_c)|,$$

against the Candelas–de la Ossa–Green–Parkes conifold-point mirror period and the Ooguri–Strominger–Vafa attractor rate $5\log 5$. Item (6) states the non-perturbative completion target for BCOV at the quintic through genus 22 at each special locus and names the exact remaining obstruction: a global resurgent sheaf across Stokes walls matching BCOV monodromy on the

full Kahler moduli space. Items (3), (6), and (7) remain conjectural, sharpened against CDGP period data and HKQ BPS data.